MELODY INFILLING WITH USER-PROVIDED STRUCTURAL CONTEXT

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ABSTRACT

This paper proposes a novel Transformer-based model for music score infilling, to generate a music passage that fills in the gap between given past and future contexts. While existing infilling approaches can generate a passage that connects smoothly locally with the given contexts, they do not take into account the musical form or structure of the music and may therefore generate overly smooth results. To address this issue, we propose a structure-aware conditioning approach that employs a novel attention-selecting module to supply user-provided structure-related information to the Transformer for infilling. With both objective and subjective evaluations, we show that the proposed model can harness the structural information effectively and generate melodies in the style of pop of higher quality than the two existing structure-agnostic infilling models.

1. INTRODUCTION

In recent years, machine learning techniques have been widely applied to symbolic music generation. A large number of models attain *sequential generation* by accounting for only the *past context*, i.e., the generated music depends on only the preceding musical content [1–14]. While sequential generation can find useful use cases, it does not align with typical human compositional practices which can be non-sequential in nature. Musicians often write motifs or small pieces to get inspiration first, before working on the middle parts to connect them.

Hence, we focus on the scenario when both the past and *future* contexts are given, which is called *music score infilling* or inpainting [15]. As shown in Figure 1(a), the task is to let models fill in the missing part between the two given segments. Prompt-based conditioning approaches [15–23] have been applied to such a task in recent years, treating the two given segments as the "prompt." Among them, the variable-length infilling model (VLI) [20] obtains promising results by adding special positional encodings to XLNet [24], a permutation-based language model that is naturally suitable for generative tasks with given bi-

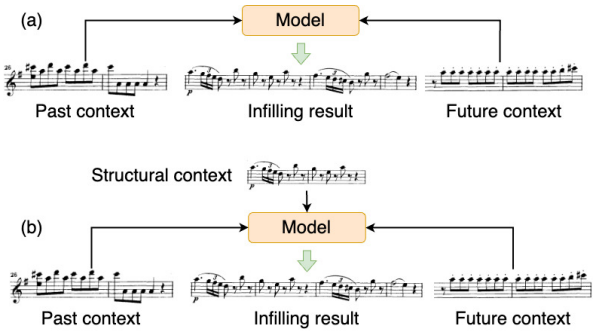


Figure 1: Comparison between (a) structure-agnostic and (b) structure-aware approaches for music score infilling.

directional contexts. The experiment of VLI shows that their model is capable of connecting the past and future contexts smoothly locally for infilling solo piano passages of up to 4 bars (measures).

Considering composers usually write musical pieces in a hierarchical manner [25], we note that prompt-based conditioning approaches have a strong limitation: they generate results with only consideration of local smoothness among the past context, future context, and result, without taking care of the overall musical *form* or *structure* of the music. For instance, a composer may like to write a song in a musical form of `ABA′B′`. If we consider the concatenation of the segments corresponding to `A` and `B` (i.e., `AB`) as the past context, and the segment corresponding to `B′` as the future context, and feed them to an existing infilling model, the model may generate a sequence that consists of similar melody and chord progression as the segments corresponding to `B` and `B′`, not the intended repetition or variation of the segment corresponding to `A`.

To address this issue, we propose in this paper a novel **structure-aware** setting for music infilling. As shown in Figure 1(b), besides the past and future contexts exploited by conventional structure-agnostic, prompt-based models, out approach additionally capitalizes for the infilling task the *structural context*, a music segment corresponding to a certain part of the whole music that is supposed to share the same *structure label* (such as `A` or `B`) with the missing segment. Accordingly, besides local smoothness, the model also needs to consider the similarity between the infilled segment and the structural context. Here, we assume the structural context is provided by a user, not generated by a model. For example, the user may designate the segment corresponding to `A` as the structural context, thereby

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inform the model with the intended musical form.

We improve upon the VLI model [20] in the following ways to realize structure-aware infilling. First, we use the classic Transformer [26–28] instead of the more sophisticated XLNet [24] as the model backbone, to make it easier to add a conditioning module to exploit the structural context. To improve the capability of the Transformer to account for bi-directional contexts, we propose two novel components, the *bar-count-down technique* (Section 3.2) and *order embeddings* (Section 3.3), which respectively give the model an explicit control of the length of the generated music, and a convenient way to attend to the future context. Second, being inspired by the Theme Transformer [29], we use not a Transformer decoder-only architecture but a sequence-to-sequence (seq2seq) Transformer encoder/decoder architecture, using the cross-attention between the encoder and decoder as the conditioning module to account for the structural context. Moreover, we propose an *attention-selecting module* that allows the Transformer to access multiple structural contexts while infilling different parts of a music piece, which can be useful both in the training and inference time (Section 3.4) .

For evaluation, we compare our model with two strong baselines, the VLI [20] and the work of Hsu & Chang [21], on the task of symbolic-domain melody infilling of 4-bar content using the POP909 dataset [30] and the associated structural labels from Dai *et al.* [31]. With objective and subjective analyses, we show that our model greatly outperforms the baselines in the structure completeness of the generated pieces, without degrading local smoothness.

We set up a webpage for demos [1] and open source our code at a public GitHub repository. [2]

2. RELATED WORK

Generating missing parts with given surrounding contexts has been attempted by early works. DeepBach [17] predicts missing notes based on the notes around them. They use two recurrent neural networks (RNNs) to capture the past and future contexts, and a feedforward neural network to capture the current context from notes with the same temporal position as the target note. COCONET [16] trains a convolutional neural network (CNN) to complete partial musical scores and explores the use of blocked Gibbs sampling as an analog to rewriting. They encode the music data with the piano roll representation and treat that as a fixed-size image, so the model can only perform fixed-length music infilling. Inpainting Net [15] uses an RNN to integrate the temporal information from a variational auto-encoder (VAE) [32] for bar-wise generation, Wei *et al.* [23] build the model with a similar concept as Inpainting Net and use the contrastive loss [33, 34] for training to improve the infilling quality. Some Transformer-based models have also been proposed to achieve music infilling. Ippolito *et al.* [18] concatenate the past and future context with a special separator token. They keep the original positional encoding of the contexts and the missing segment, which again limits the length of given contexts and generated sequence to be fixed. We see that these infilling models impose some data assumptions and thereby have certain restrictions, e.g., the length of the input sequence cannot be arbitrary, or the missing segment needs to be complete bars. The work of Hsu & Chang [21] is free of these restrictions. They use two Transformer encoders to capture the past and future context respectively and generate results with a Transformer decoder. The VLI model [20] can also realize variable-length infilling. However, to our best knowledge, no existing models have explicitly considered structure-related information for infilling.

Structure-based conditioning has been explored only recently by Shi *et al.* [29] in their Theme Transformer model for sequential music generation. They use a seq2seq Transformer to account for not only the past context but also an additional pre-given theme segment that is supposed to manifest itself multiple times in the model's generation result. The present work can be considered as an extension of their work to the scenario of music infilling.

3. METHODOLOGY

Given a past context C_{past} and a future context C_{future}, the general, **structure-agnostic music infilling** task entails generating an *infilled segment* T that interconnects C_{past} and C_{future} smoothly, preferably in a musically meaningful way. When using an autoregressive generative model such as the Transformer as the model backbone, the training object is to maximize the following likelihood function:

$$\prod_{0<k\leq|T|} P(t_k|t_{<k}, C_{\text{past}}, C_{\text{future}}) \,, \tag{1}$$

where t_k denotes the element of T at timestep k, $t_{<k}$ the subsequence consisting of all the previously generated elements, and $|\cdot|$ the length of a sequence.

Extending from Eq. (1), we propose and study in this paper a special case, called **structure-aware music infilling**, where an additional segment G representing the *structural context* is given, leading to the new objective:

$$\prod_{0<k\leq|T|} P(t_k|t_{<k}, C_{\text{past}}, C_{\text{future}}; G) \,. \tag{2}$$

As depicted in Figure 2(a), our model is based on Transformer with the encoder-decoder architecture. It uses the decoder to self-attend to the prompt (i.e., C_{past} and C_{future}) and the previously-generated elements (i.e., $t_{<k}$), and the encoder to cross-attend to the structural context G. We provide details of the proposed model below.

Note that we do not require the length of all the involved segments to be fixed; namely $|T|$, $|C_{\text{past}}|$, $|C_{\text{future}}|$ and $|G|$ are all variables in our setting.

3.1 REMI-based Token Representation

To incorporate structure-related information to our representation of the music data, we devise an extension of the REMI-based representation [8] that comprises five types

[1] https://tanchihpin0517.github.io/structure-aware_infilling
[2] https://github.com/tanchihpin0517/structure-aware_infilling

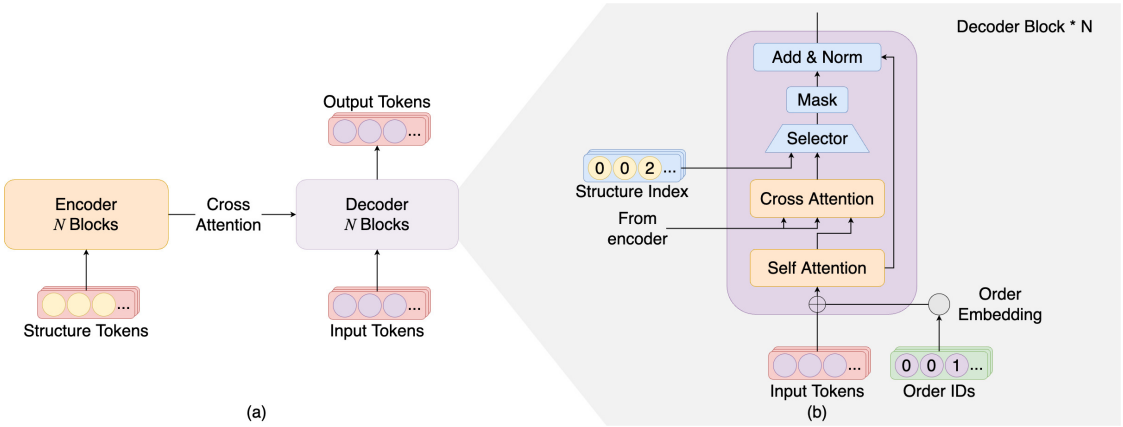


Figure 2: Schematic diagram of (a) the proposed seq2seq Transformer model for structure-aware infilling, and (b) a zoom-in of the decoder highlighting how the decoder utilizes the *order embedding* and *structure index*.

Token type	Voc. size	Values
`Bar`	32	1, 2, ... , 32
`Struct`	16	0, 1, ... , 15
`Tempo`	47	28, 32, ... , 212
`Position`	16	0, 1, ... , 15
`Pitch`	86	22, 23, ... , 107
`Duration`	16	1, 2, ... , 16

Table 1: The vocabulary of the token representation.

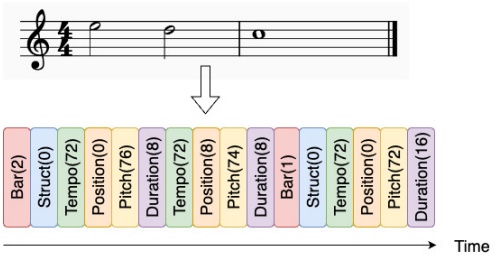


Figure 3: An example of representing a two-bar segment with our adapted REMI-based token representation. With the bar-count-down technique, `Bar(2)` and `Bar(1)` are used for the first and second `Bar` tokens, respectively.

of tokens: `Bar`, `Struct`, `Tempo`, `Position`, `Pitch` and `Duration`. Table 1 lists the vocabulary of our token representation. `Bar` consists of numbers from 1 to 32, standing for the number of remaining bars on the generation process. A music form, e.g., `ABA′B′`, consists of multiple groups of similar phrases or sections (each of multiple bars), e.g., {`A`, `A′`} and {`B`, `B′`}, where each group can be said to be associated with the same *structure label*. We use `Struct` to indicate the structure label for each bar. `Tempo` and `Position` are related to the musical metre. `Tempo` is the current tempo of beats per minute (BPM), and `Position` is the temporal distance between the onset of a musical note and the beginning of its bar, denoted by the number of 16-th notes. `Pitch` and `Duration` are related to musical notes, which are the MIDI pitch number and duration in 16-th notes, respectively. We show an example of how we encode the musical content in Figure 3.

3.2 Bar-Count-Down Technique

The work of Hsu & Chang [21] uses special `BOS` and `EOS` tokens as the "signal" to start or stop the generation process. At inference time, the infilled segment generated by their model comes to an end when the model generates an `EOS` token. While this may work fine in certain cases, doing so cannot give us an explicit control of the number of bars to be generated for the infilled segment. Such a control is preferable when we want to make sure that the previous context and the future context are a certain number of bars apart, which is highly needed for structure-aware infilling. For example, a user may want to specify the music form as `A8B8A′8B′8`, meaning that all the four sections `A`, `B`, `A′`, `B′` are eight-bar long each.

To have such a control, we employ a special token representation technique called "bar-count-down." For each infilled sequence T, we adjust the suffix number of `Bar` tokens to match the length of T. Take Figure 3 as an example: The number in `Bar` tokens are counted down from two because the sequence in Figure 3 is 2-bars long. Once training a model with this setup, the length of T can be controlled effectively by the number of remaining bars associated with the first `Bar` token given on generation.

3.3 Order Embedding

Transformers with causal masking are mainly designed for sequential generation. To apply the model to infilling tasks where the missing part is in the middle of the input sequence, the model proposed by Hsu & Chang [21] attends to bi-directional context from two encoders with cross-attention. As we want to instead use only the self-attention of the decoder to exploit bi-directional information, we reorder the sequence $\{C_{\text{past}}, T, C_{\text{future}}\}$ to $\{C_{\text{past}}, C_{\text{future}}, T\}$. Doing so would however change the original positional relationship among C_{past}, T, and C_{future}.[3] As depicted

[3] The principal idea of the attention mechanism [26] is to use the token embeddings of two tokens to compute their correlation, leading to the so-

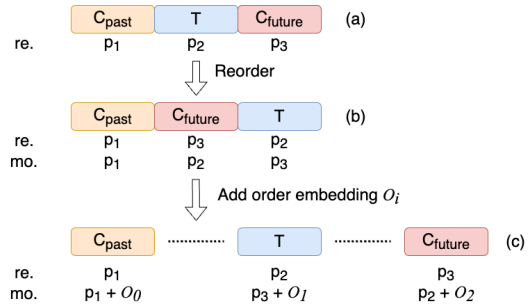


Figure 4: Illustration of the order embedding, where $re.$ and $mo.$ stand for the *real position* and *model-viewed position*. Conceptually, we shift the model-viewed position by O_i to make it consistent with the real position.

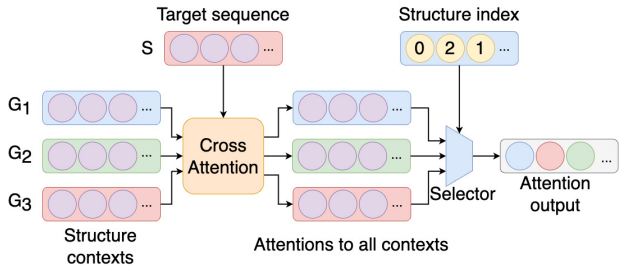


Figure 5: The schematic of attention selecting. Multiple structural contexts G_1, G_2, G_3 are passed to the cross-attention block simultaneously, and the output is filtered by the *selector* with the *structure index*.

in Figure 4(b), the *real* positional relationships among the segments entails associating the tokens in T with positional embeddings corresponding to a set of positions p_2 that signifies the model the segment T is after C_{past} (i.e., $p_1 \prec p_2$) and is before C_{future} ($p_2 \prec p_3$).[4] After reordering, however, using the typical way of computing the positional embeddings from left to right by the Transformers, T would be assigned with positional embeddings corresponding to p_3, meaning that by the *model-viewed* positional relationships, T is after both C_{past} and C_{future}. The real and the model-viewed are mismatched.

We introduce a new position-related *segment embedding* called "order embedding" to tackle this issue. Specifically, for the positional embeddings for all the tokens in C_{past}, we add to them the same additional embedding corresponding to a positional "offset" or "order," denoted as O_0. We similarly incorporate O_1 and O_2 to the positional embeddings for the tokens in T and C_{future}. As long as the "offset" is big enough, we can have $p_1 + O_0 \prec p_3 + O_1 \prec p_2 + O_2$, ensuring that the real and model-viewed positional relationships are matched, as depicted in Figure 4(c).

We note that, as the same order embedding is added to the positional embeddings of all the tokens in a segment, the proposed idea works nicely regardless of whether the segment lengths $|C_{\text{past}}|$, $|T|$, $|C_{\text{future}}|$ are fixed or not.

3.4 Attention-Selecting Module

While the formulation of Eq. (2) considers only one structural context G, in practice, we may want to designate multiple structural contexts $\{G_1, G_2, \ldots, G_N\}$ for infilling, with each segment G_n corresponding to a certain structure label such as `A` and `B`. This is useful, for example, when the first part of the infilled segment T is meant to be similar to phrase `A`, while the latter part similar to phrase `B`. To indicate which G_n a specific token t_k of T should refer to, we define the *structure index* $y_k \in \{0, \ldots, N\}$, for $k = \{1, \ldots, |T|\}$. The structure indices are also given by the user when the user specifies the intended musical form. With all these, we extend Eq. (2) to:

$$\prod_{0<k\leq|T|} P(t_k|t_{<k}, C_{\text{past}}, C_{\text{future}}; G_1, ..., G_N, y_k)\,. \quad (3)$$

We use $y_k = 0$ to indicate the case where t_k is supposed to follow none of the structural contexts (e.g., when t_k is part of the bridge). When $y_k \neq 0$, t_k follows only one of the structural context G_{y_k}. In our implementation, for tokens whose $y_k = 0$, we only use the self-attention to attend to the prompt and $t_{<k}$, leading to a formulation akin to Eq. (1). When $y_k \neq 0$, we have a "selector" that picks by y_k the structural context G_{y_k} to be attended to via cross-attention, resulting in a formulation akin to Eq. (2). This is depicted in Figure 2(b) and Figure 5.

Figure 5 also shows that, in our implementation, instead of computing the cross attention between the target sequence with a specific structural context G_{y_k}, we actually compute the cross attention between the target sequence with every structural contexts $\{G_1, G_2, \ldots, G_N\}$ and let the selector pick the right one. While this seems a waste of computing, we find doing so faster when GPUs are used.[5]

4. EXPERIMENTAL SETUP

We collect the melody data[6] from the POP909 dataset compiled by Wang *et al.* [30]. POP909 contains 909 MIDI files of pop piano performances, though we discard 8 of them due to errors encountered in the preprocessing stage. We split all remaining songs into 811 (90%) songs for training and 91 (10%) songs for testing. The 16-th note is set as the minimal temporal resolution to quantize the tempo, beat, and duration for reducing the vocabulary size.

Dai *et al.* [31] publicly share the *structural information* of songs in POP909 with letters and integers to indicate the structure labels and their lengths in bars, such as

`i4 A8 B8 x4 A8 B8 B8 X2 c4 c4 X2 B9 o2`, (4)

called attention score. However, using the token embeddings alone fails to consider the position-related relations of the two tokens (e.g., whether they are neighbors or distant apart). Accordingly, in practice, people add token-wise positional embeddings to the token embeddings before computing their attention [35–37].

[4] We use $p \prec q$ to denote that any elements in the set p is smaller, or "temporally before," any elements in the set q.

[5] There is another implementation detail: actually, not only the tokens in T but also those in C_{past} and C_{future} would go through the Transformer's self- and cross-attention blocks. This is to get the latent vectors for the tokens in C_{past} and C_{future}. In doing so, we calculate and employ the structure indices for C_{past} and C_{future} as well.

[6] Please note that the "melody" data in this paper is not exact monophonic music. We merge two midi tracks, "MELODY" and "BRIDGE" of the MIDI files of POP909 [30] to generate the melody data, which is monophonic most of the time but not always.

where the music phrases labeled with the same letter are considered to share the same structural context.[7] [8] Besides, the musical phrases with the labels `i`, `x`, and `o` are respectively the introduction, bridge, and ending, which do not have structural context in our setting. For each song, we use the phrases corresponding to the first occurrence of the structure labels such as `A` and `B` as the structural contexts G_1, G_2, etc. For example, we choose bars 5–12 (i.e., the `A8` phrase after `i4`) and bars 13–20 (i.e., the `B8` phrase before `x4`) in the example shown in Eq. (4) as the structural contexts for `A` and `B` since they are the first music phrases in the song with the corresponding labels.

The training data is generated with the following steps: (i) iterating through all labels except for the first and last ones in the structural information, (ii) choosing their corresponding music phrases as the infilled sequences T, and (iii) concatenating T with their preceding and following 6-bars music segments to get $\{C_{\text{past}}, T, C_{\text{future}}\}$, yielding 8,607 data in total. Before feeding the training data into the model, we reorder the input sequences and insert additional special tokens, `BOS`, `SEP`, and `EOS`, to change them into $\{\texttt{BOS}, C_{\text{past}}, \texttt{SEP}, C_{\text{future}}, \texttt{SEP}, T, \texttt{EOS}\}$. The reordered input sequence and structural contexts are transformed to the embeddings with size 512. The model consists of an encoder and a decoder, where both of them consist of six 8-head self-attention layers with intermediate layers of dimension 2,048. Each layer of the encoder and decoder is connected with an 8-head cross-attention, as shown in Figure 2. The output from the model is transformed back to a probability distribution of the vocabulary with the softmax function. At inference time, we use nucleus sampling [38] to sample the output tokens with the threshold value of 0.9.

For evaluation, we create the testing data by: (i) searching from the testing songs all the 4-bar phrases that correspond to only a structure label, (ii) keeping only the phrases that share the same structure label with one of its two neighboring phrases but a different structure label with the other neighbor, and (iii) setting the 4-bar phrase as the target T and concatenate them with their preceding and succeeding 6-bar music segments, which are set as the past context C_{past} and future context C_{future}, respectively. We get 156 test cases of $\{C_{\text{past}}, T, C_{\text{future}}\}$, each with 16 bars (i.e., $6+4+6$). All the cases have the form of, e.g., `AA'B` or `ABB'`, and the target lengths are **4 bars** with arbitrary number of notes (hence *variable* sequence lengths). In this setting, the attention selection mechanism is used only for training, since the target sequence T in our testing data only have one structural context to refer to.

We consider VLI [20] and the model from Hsu & Chang [21] as the baselines. By design, only our model has access to an external structural context. However, we consider the comparison as valid, since C_{past} and C_{future} are presumably long enough to provide sufficient context, and at least one of them has the same structure label as T. Besides, in our implementation, we found the model Hsu & Chang [21] rarely generates infilled segments with the desirable number of bars. Therefore, we slightly improve their model by incorporating the bar-count-down technique.

[7] The lower-case and capital letters indicate respectively *non-melodic* and *melodic* phrases (i.e., where a clear melody is present, mostly a vocal line or an instrument solo), but we do not use this information.

[8] Depending on the underlying musical form of a song, different songs may have different numbers of phrase groups. For example, a song with a simpler form may only have phrase groups `A` and `B`, while other songs have much more. In the POP909 dataset, a song can contain up to 9 unique phrase groups, which are labeled as `A`, `B`, `C`, `D`, etc.

	$H \downarrow$	$GS \uparrow$	$D \downarrow$
Ours	**2.75**$\pm$0.80	**0.70**$\pm$0.08	**25.73**$\pm$19.45
VLI [20]	3.47$\pm$1.57	0.67$\pm$0.09	49.40$\pm$25.12
Hsu [21]	9.87$\pm$4.64	0.64$\pm$0.09	65.41$\pm$38.00
Original	2.78$\pm$0.89	0.70$\pm$0.09	0.00$\pm$ 0.00

Table 2: Objective evaluation results. H: pitch class histogram cross entropy, GS: grooving pattern similarity, D: melody distance (↑/↓: the higher/lower the better).

5. OBJECTIVE EVALUATION RESULTS

We propose three new metrics for objective evaluation, all of which have not been used in the literature of music infilling. The first two metrics, **pitch class histogram cross entropy** (H) and **grooving pattern similarity** (GS), are extensions of the ones proposed by Wu & Yang for sequential generation [10] to our infilling task, evaluating respectively the consistency in terms of pitch class distribution (which is related to tonality) and rhythmic pattern. For H, we compute per test case the pitch class histogram of T, and that histogram of the concatenation of C_{past} and C_{future}, and then report the cross entropy between these two histograms. For GS, we use per bar a 16-dim binary vector indicating where there is at least a note onset for every position in a bar, calculate one minus the normalized XOR difference between every pair of bars [10], one from T and the other from either C_{past} and C_{future}, and then report the average per test case. The third metric, **melody distance** (D) measures the melody distance (dissimilarity) between the infilled segment T and the ground truth one (denoted as $T\#$) using the algorithm proposed by Hu *et al.* [39]. Lower H and higher GS may imply that T connects C_{past} and C_{future} smoothly, while lower D indicates that the generation result is similar to a human-made one.

Table 2 shows that our model achieves the best result in all the three metrics, followed by VLI and then the model of Hsu & Chang. Besides being consistent with the contexts, the infilling result of our model is closest to the ground truth one $T\#$, demonstrating the effectiveness of exploiting the structural context. Figure 6 exemplifies how the infilled bars by our model fit the desired musical form.

6. SUBJECTIVE EVALUATION RESULTS

We conduct additionally an online user study for subjective evaluation. We have 91 anonymous volunteers, where 12 of them are marked as professionals according to the question about their musical background. Each subject is pre-



Figure 6: Results generated by (a) our model and (b) VLI. The structural relations of bars 4-5 (green area) and bars 8-9 (red area) are preserved well in (a) but not in (b).

		M	R	S	O
all	Ours	**3.46**	**3.51**	**3.40**	**3.42**
	VLI [20]	2.96	3.14	3.12	2.97
	Hsu [21]	2.60	2.95	2.75	2.64
	Real	**3.77**	**3.77**	**3.62**	**3.66**
pro	Ours	**3.58**	**3.28**	**3.28**	**3.42**
	VLI [20]	2.67	2.86	2.78	2.72
	Hsu [21]	2.36	2.75	2.39	2.44
	Real	**3.61**	**3.56**	**3.42**	**3.42**

Table 3: Results of the user study: mean opinion scores in 1–5 in **M** (melodic fluency), **R** (rhythmic fluency), **S** (Structureness), and **O** (Overall), from 'all' the participants or only the music 'pro'-fessionals.

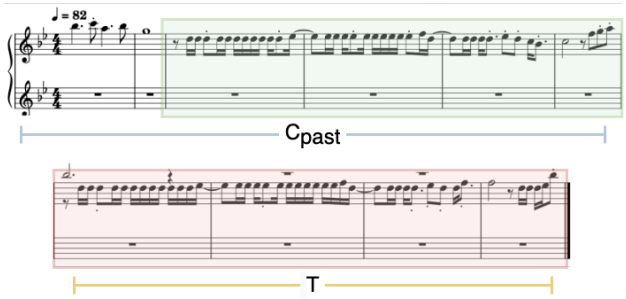


Figure 7: A result generated by our model, where C_{future} is not shown in this figure. Bars 3–6 (green area) and bars 7–10 (red area) look identical since the model overly imitates the given structural context. (Best viewed in color.)

sented with 3 out of 15 sets of music segments randomly sampled from the testing data. We inform them that the first and last 6 bars are the given prompts, and the middle 4 bars are the music generated by a model. Each set of music contains in random order 4 music including 3 generated by the models and 1 from the real data. The subjects rate each of the 4 music in a 5-point Likert scale (the higher the better) according to their (i) **melodic fluency**: do the pitches of notes go in the right tonality and connect the contexts fluently? (ii) **rhythmic fluency**: are the notes played on the right beats? (iii) **structureness**: how is the generated part of the music similar to its contexts? (ix) **overall**: how much do they like the music?

Table 3 shows the mean opinion scores (MOS) of the user study. Echoing the result of the objective evaluation, the proposed model outscores the baselines by a large margin in all the four subjective metrics, and is close to the real music with a small gap. The same observations can be made from either the average result of all the subjects, or only that from the 12 professionals.

However, we notice that our model may overly imitate the given structural contexts in some cases, as exemplified in Figure 7. When this happens, the generated music sounds rigid and non-creative. We conjecture that this can be attributed to the limited diversity of our training data—the melodies corresponding to the same structure label in POP909 appear to be too similar to each other, which may not be uncommon for pop songs. To study this, we implement additionally a 'Copy' baseline that simply copies the structural contexts as the result, and include its infilling result to the demo website. Our own subjective listening of its result confirms that the proposed method still outperforms the 'Copy' baseline most of the time, as the connections between the targets and their contexts are considered by our model, but not by the 'Copy' baseline.

7. CONCLUSION

In this paper, we have proposed a new structure-aware conditioning approach for music score infilling. To help Transformers exploit bi-directional contexts, we employ the order embedding to shift the position viewed by the model. Besides, we introduce a new attention-selecting mechanism to account for multiple structural contexts. Evaluations on 4-bar melody infilling validate the superiority of the proposed model over two existing Transformer-based structure-agnostic infilling methods [20, 21].

We can extend this work in three ways. First, instead of relying on user inputs, we may build models that generate the musical form automatically and predict the structural context for a specific infilled segment to refer to. Second, we like to expand our work to other music genres and polyphonic music. Finally, we like to study how the attention-selecting mechanism can be applied to sequential generative tasks such as theme-based generation [29].

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9. REFERENCES

[1] N. Boulanger-Lewandowski, Y. Bengio, and P. Vincent, "Modeling temporal dependencies in high-dimensional sequences: Application to polyphonic music generation and transcription," in *Proc. International Conference on Machine Learning*, 2012, pp. 1881–1888.

[2] F. Colombo, "Algorithmic composition of melodies with deep recurrent neural networks," *Poster Session of International Conference on Artificial Intelligence and Statistics*, 2016.

[3] B. L. Sturm, J. F. Santos, O. Ben-Tal, and I. Korshunova, "Music transcription modelling and composition using deep learning," in *Proc. Conference on Computer Simulation of Musical Creativity*, 2016.

[4] A. Roberts, J. Engel, C. Raffel, C. Hawthorne, and D. Eck, "A hierarchical latent vector model for learning long-term structure in music," in *Proc. International Conference on Machine Learning*, 2018, pp. 4364–4373.

[5] L.-C. Yang, S.-Y. Chou, and Y.-H. Yang, "MidiNet: A convolutional generative adversarial network for symbolic-domain music generation," in *Proc. International Conference on Music Information Retrieval*, 2017.

[6] C.-Z. A. Huang, A. Vaswani, J. Uszkoreit, N. Shazeer, I. Simon, C. Hawthorne, A. M. Dai, M. D. Hoffman, M. Dinculescu, and D. Eck, "Music Transformer," in *Proc. International Conference on Learning Representations*, 2019.

[7] C. Donahue, H. H. Mao, Y. E. Li, G. W. Cottrell, and J. McAuley, "LakhNES: Improving multi-instrumental music generation with cross-domain pre-training," in *Proc. International Conference on Music Information Retrieval*, 2019.

[8] Y.-S. Huang and Y.-H. Yang, "Pop Music Transformer: Beat-based modeling and generation of expressive pop piano compositions," in *Proc. ACM International Conference on Multimedia*, 2020, pp. 1180–1188.

[9] Y. Ren, J. He, X. Tan, T. Qin, Z. Zhao, and T.-Y. Liu, "PopMAG: Pop music accompaniment generation," in *Proc. ACM Multimedia*, 2020.

[10] S.-L. Wu and Y.-H. Yang, "The Jazz Transformerm on the front line: Exploring the shortcomings of ai-composed music through quantitative measures," in *Proc. International Conference on Music Information Retrieval*, 2020.

[11] S. Dai, Z. Jin, C. Gomes, and R. B. Dannenberg, "Controllable deep melody generation via hierarchical music structure representation," in *Proc. International Conference on Music Information Retrieval*, 2021.

[12] S.-L. Wu and Y.-H. Yang, "MuseMorphose: Full-song and fine-grained music style transfer with one Transformer VAE," *arXiv preprint arXiv:2105.04090*, 2021.

[13] J. Liu, Y. Dong, Z. Cheng, X. Zhang, X. Li, F. Yu, and M. Sun, "Symphony generation with permutation invariant language model," *arXiv preprint arXiv:2205.05448*, 2022.

[14] H.-W. Dong, K. Chen, S. Dubnov, J. McAuley, and T. Berg-Kirkpatrick, "Multitrack Music Transformer: Learning long-term dependencies in music with diverse instruments," *arXiv preprint arXiv:2207.06983*, 2022.

[15] A. Pati, A. Lerch, and G. Hadjeres, "Learning to traverse latent spaces for musical score inpainting," in *Proc. International Conference on Music Information Retrieval*, 2019.

[16] C.-Z. A. Huang, T. Cooijmans, A. Roberts, A. Courville, and D. Eck, "Counterpoint by convolution," in *Proc. International Society for Music Information Retrieval*, 2017.

[17] G. Hadjeres, F. Pachet, and F. Nielsen, "Deepbach: a steerable model for bach chorales generation," in *Proc. International Conference on Machine Learning*, 2017, pp. 1362–1371.

[18] D. Ippolito, A. Huang, C. Hawthorne, and D. Eck, "Infilling piano performances," in *Proc. NIPS Workshop on Machine Learning for Creativity and Design*, 2018.

[19] T. Bazin and G. Hadjeres, "Nonoto: A model-agnostic web interface for interactive music composition by inpainting," in *Proc. International Conference on Computational Creativity*, 2019.

[20] C.-J. Chang, C.-Y. Lee, and Y.-H. Yang, "Variable-length music score infilling via XLNet and musically specialized positional encoding," in *Proc. International Conference on Music Information Retrieval*, 2021.

[21] J.-L. Hsu and S.-J. Chang, "Generating music transition by using a Transformer-based model," *Electronics*, vol. 10, no. 18, 2021.

[22] C.-P. Tan, C.-J. Chang, A. W. Y. Su, and Y.-H. Yang, "Music score expansion with variable-length infilling," in *Proc. International Conference on Music Information Retrieval*, 2021, late-breaking and demo paper.

[23] S. Wei, G. Xia, Y. Zhang, L. Lin, and W. Gao, "Music phrase inpainting using long-term representation and contrastive loss," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022, pp. 186–190.

[24] Z. Yang, Z. Dai, Y. Yang, J. Carbonell, R. R. Salakhutdinov, and Q. V. Le, "XLNet: Generalized autoregressive pretraining for language understanding," *Proc. Advances in Neural Information Processing Systems*, vol. 32, 2019.

[25] F. Lerdahl and R. S. Jackendoff, *A Generative Theory of Tonal Music, Reissue, with a New Preface*. MIT press, 1996.

[26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Proc. Advances in Neural Information Processing Systems*, vol. 30, 2017.

[27] Z. Dai, Z. Yang, Y. Yang, J. Carbonell, Q. V. Le, and R. Salakhutdinov, "Transformer-XL: Attentive language models beyond a fixed-length context," in *Proc. Annual Meeting of the Association for Computational*, 2019.

[28] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," 2018.

[29] Y.-J. Shih, S.-L. Wu, F. Zalkow, M. Muller, and Y.-H. Yang, "Theme Transformer: Symbolic music generation with theme-conditioned transformer," *IEEE Transactions on Multimedia*, 2022.

[30] Z. Wang, K. Chen, J. Jiang, Y. Zhang, M. Xu, S. Dai, G. Bin, and G. Xia, "POP909: A pop-song dataset for music arrangement generation," in *Proc. International Conference on Music Information Retrieval*, 2020.

[31] S. Dai, H. Zhang, and R. B. Dannenberg, "Automatic analysis and influence of hierarchical structure on melody, rhythm and harmony in popular music," in *Proc. Conference on AI Music Creativity*, 2020.

[32] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *Proc. International Conference on Learning Representations*, 2014.

[33] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. International conference on machine learning*, 2020, pp. 1597–1607.

[34] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 9729–9738.

[35] P. Shaw, J. Uszkoreit, and A. Vaswani, "Self-attention with relative position representations," in *Proc. Conf. North American Chapter of the Association for Computational Linguistics*, 2018.

[36] G. Ke, D. He, and T.-Y. Liu, "Rethinking positional encoding in language pre-training," in *Proc. International Conference on Learning Representations*, 2021.

[37] A. Liutkus, O. Cífka, S.-L. Wu, U. Simsekli, Y.-H. Yang, and G. Richard, "Relative positional encoding for Transformers with linear complexity," in *Proc. International Conference on Machine Learning*, 2021.

[38] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, "The curious case of neural text degeneration," in *Proc. International Conference on Learning Representations*, 2020.

[39] N. Hu, R. B. Dannenberg, and A. L. Lewis, "A probabilistic model of melodic similarity," in *Proc. International Computer Music Conference*, 2002.